

Grace Zhang

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TECHNICAL SKILLS

Languages: Python, Go, TypeScript, JavaScript, Java, C/C++, SQL

Frameworks & Libraries: Django, React, Node.js, Express, Playwright, Tailwind CSS

Developer Tools: AWS, GCP, Docker, Git, Linux, PostgreSQL, MySQL

EXPERIENCE

Software Engineer Intern

Jan. 2026 – Apr. 2026

Mantle Equity

Toronto, ON

- Built a **Go** service that analyzes **Playwright** trace artifacts from **AWS S3** using **Gemini (Vertex AI)** to generate automated root cause reports, reducing debugging time from **20 minutes to seconds**
- Integrated **Vertex AI** into a **Go-based** financial data pipeline to reconcile duplicate investment holdings, using identifier-based validation to eliminate **95%** of duplicate records
- Automated investor portal workflows using **Playwright** with **TypeScript**, handling MFA, dynamic table parsing, and document downloads to support daily ingestion pipelines

Software Developer Intern

May 2025 – Aug. 2025

Empire Life

Kingston, ON

- Built the withdrawals step of a high-visibility advisor wizard in **React/TypeScript** with bilingual validation, enabling accurate calculations and unblocking product testing
- Enhanced a production **Django** web app serving thousands of advisors across Canada by implementing new features and resolving production issues
- Reduced search latency by **40%** by implementing fuzzy search with **Fuse.js** in **Django**

Web Developer Intern

Sept. 2024 – Dec. 2024

University Health Network

Toronto, ON

- Migrated the UHN Research website to **dotCMS**, building reusable templates and components to streamline page creation
- Engineered reusable UI components in **TypeScript** for the UHN Research intranet and public websites, improving responsiveness and maintainability
- Reduced manual content updates by **30%** by implementing dynamic content workflows using **dotCMS APIs** and **Velocity**

Software Developer Intern

Jan. 2024 – Apr. 2024

Pratt & Whitney

Mississauga, ON

- Built a full-stack issue tracking module using **PHP**, **JavaScript**, and **SQL** to centralize and streamline requirements management workflows
- Implemented collapsible issue rows, nested comment threads, and file uploads, reducing issue resolution time by **20%**
- Developed secure backend scripts for binary data encoding and database storage, ensuring consistent file retrieval and preventing data loss

Robotics Operations & Software Developer Intern

May 2023 – Aug. 2023

Swap Robotics

Kitchener, ON

- Integrated **AWS SES** with **Express.js** to automate email alerts for safety infractions, reducing manual effort by **80%**
- Implemented backend services with **Node.js** and **Prisma ORM** for real-time safety monitoring and KPI reporting
- Revamped a Dockerized **React** dashboard with **REST API** integrations to automate safety tracking and real-time insights

PROJECTS

Spotify Lyric Overlay App | *Electron, Node.js, Express, Spotify Web API, OAuth 2.0 PKCE*

[Link](#)

- Built a cross-platform **Electron** desktop app with a draggable overlay that displays real-time synchronized Spotify lyrics
- Implemented a **Node/Express** backend integrating the **Spotify Web API (OAuth 2.0 PKCE)** for playback tracking and **LRCLib** for lyric retrieval, parsing timestamped LRC files and caching lyrics for repeated playback

EDUCATION

University of Waterloo

Bachelor of Computer Engineering

Waterloo, ON

Expected May 2027

- **Relevant Courses:** Algorithms & Data Structures, Systems Programming, Database Systems, OS, Computer Networks